

Experiment No. 6: Use Sleuth Kit to Analyze Digital Evidence

Aim

To use the Sleuth Kit command-line tools to analyze a forensic disk image, identify the file system and partitions, list files and directories, recover a file, and analyze file metadata.

Software Used

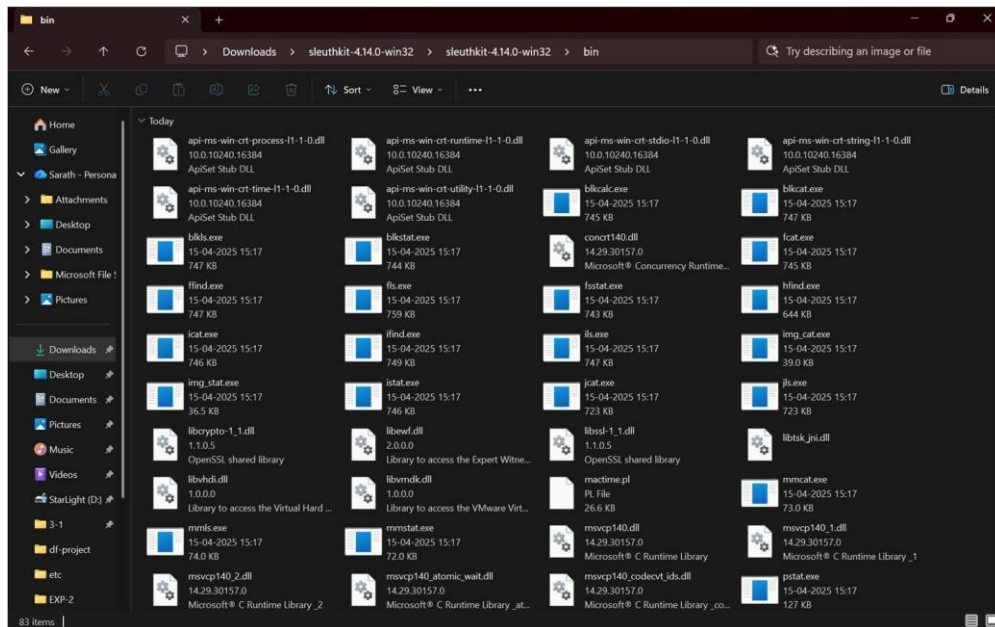
- Sleuth Kit 4.14.0
- Windows PowerShell
- Forensic Disk Image

Description

The Sleuth Kit (TSK) is a collection of command-line tools used for analyzing disk images and recovering digital evidence. In this experiment, the forensic disk image `4Dell Latitude CPi.E01` is analyzed using Sleuth Kit tools.

Step 1: Install and Open Sleuth Kit

The Sleuth Kit tools were downloaded and the bin directory was opened. The bin directory contains tools such as fsstat.exe, mmls.exe, fls.exe, icat.exe, istat.exe and img_stat.exe.



Step 2: Identify the File System Using fsstat

The fsstat command was used to identify and analyze the file system present in the forensic image

The output shows that the file system is NTFS and the image contains Windows XP file-system information.

```

Windows PowerShell
PS C:\Users\bharg\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin> .\fsstat.exe -i ewf -o 63 "D:\3-1\Digital Forensic s\EXP-6\4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
  
```

Step 3: List Partitions Using mmls

The mmls command was used to identify the partition structure of the forensic disk image.

The output shows the DOS partition table and an NTFS/exFAT partition.

Step 4: Analyze Files and Directories Using fls

```
PS C:\Users\bharg\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-  
-win32\bin> .\mmls.exe "D:\3-1\Digital Forensics\EXP-6\4Dell Latitude  
CPi.E01"  
DOS Partition Table  
Offset Sector: 0  
Units are in 512-byte sectors
```

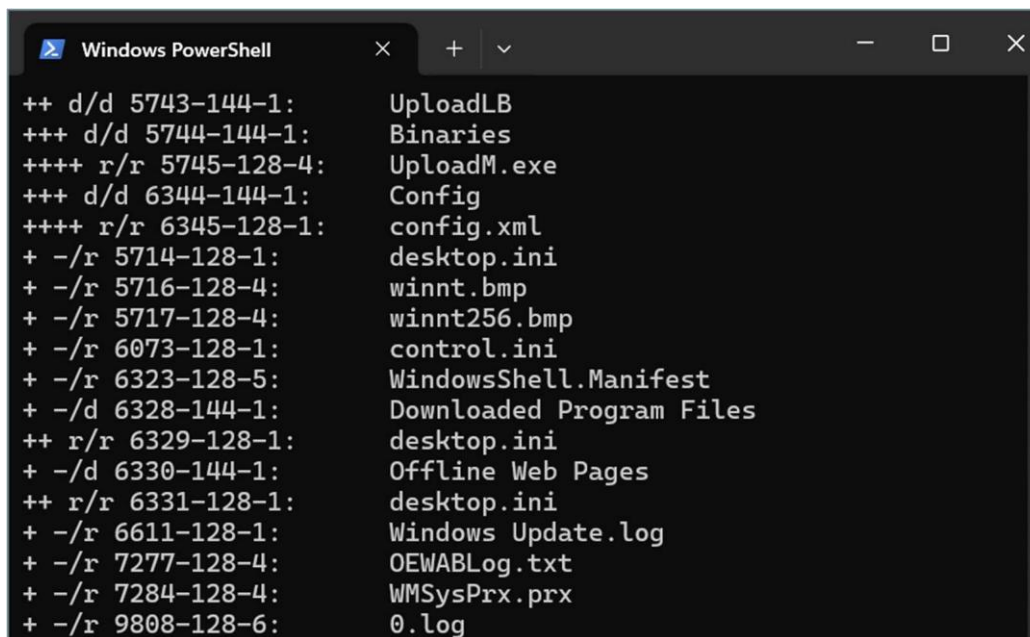
	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table
(#0)					
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT
(0x07)					
003:	-----	0009510480	0009514259	0000003780	Unallocated

The fls command was used to recursively list the files and directories contained in the forensic image.

The output was saved into file_list.txt.

```
PS C:\Users\bharg\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14  
0-win32\bin> .\fls.exe -i ewf -o 63 -r "D:\3-1\Digital Forensi  
cs\EXP-6\4Dell Latitude CPi.E01" > "D:\3-1\Digital Forensics\EXP  
-6\file_list.txt"
```

Screenshot 4 – fls command execution

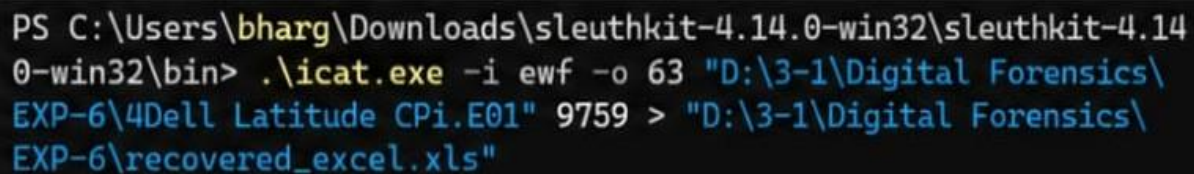


```
Windows PowerShell
++ d/d 5743-144-1: UploadLB
+++ d/d 5744-144-1: Binaries
++++ r/r 5745-128-4: UploadM.exe
+++ d/d 6344-144-1: Config
++++ r/r 6345-128-1: config.xml
+ -/r 5714-128-1: desktop.ini
+ -/r 5716-128-4: winnt.bmp
+ -/r 5717-128-4: winnt256.bmp
+ -/r 6073-128-1: control.ini
+ -/r 6323-128-5: WindowsShell.Manifest
+ -/d 6328-144-1: Downloaded Program Files
++ r/r 6329-128-1: desktop.ini
+ -/d 6330-144-1: Offline Web Pages
++ r/r 6331-128-1: desktop.ini
+ -/r 6611-128-1: Windows Update.log
+ -/r 7277-128-4: OEWABLog.txt
+ -/r 7284-128-4: WMSysPrx.prx
+ -/r 9808-128-6: 0.log
```

Step 5: Recover a File Using icat

The inode number of excel.xls was identified from the file listing. The icat command was then used to recover the file from the forensic image.

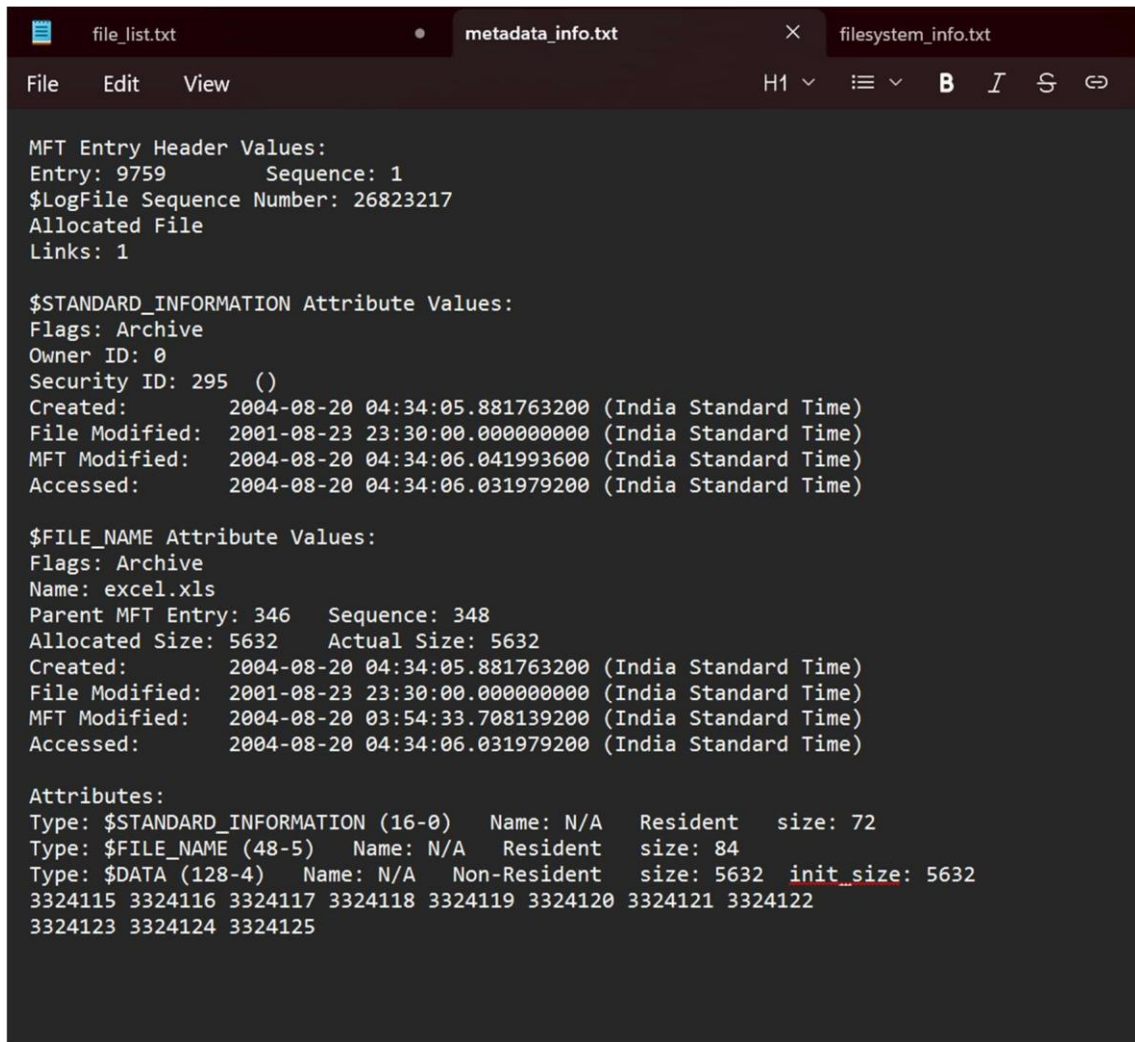
The recovered file was saved as: recovered_excel.xls



```
PS C:\Users\bharg\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14
0-win32\bin> .\icat.exe -i ewf -o 63 "D:\3-1\Digital Forensics\
EXP-6\4Dell Latitude CPi.E01" 9759 > "D:\3-1\Digital Forensics\
EXP-6\recovered_excel.xls"
```

Step 6: Analyze File Metadata Using istat

The istat command was used to analyze the metadata associated with the excel.xls file.



The screenshot shows a text editor with three tabs: 'file_list.txt', 'metadata_info.txt', and 'filesystem_info.txt'. The 'file_list.txt' tab is active, displaying detailed file metadata for 'excel.xls'. The metadata is organized into sections: MFT Entry Header Values, \$STANDARD_INFORMATION Attribute Values, \$FILE_NAME Attribute Values, and a list of Attributes.

```
file_list.txt  metadata_info.txt  filesystem_info.txt
File Edit View H1  B I S ↺

MFT Entry Header Values:
Entry: 9759      Sequence: 1
$LogFile Sequence Number: 26823217
Allocated File
Links: 1

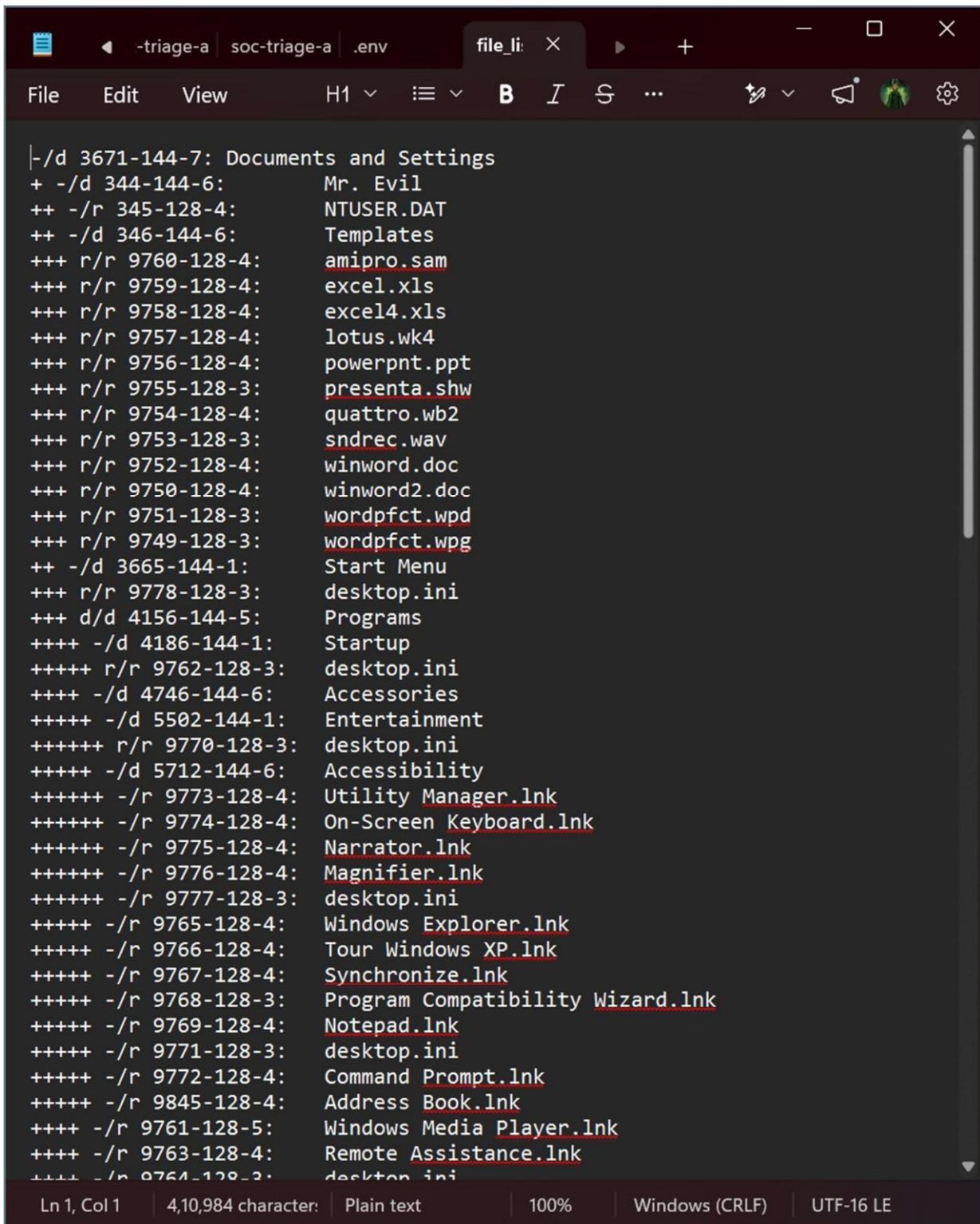
$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 295  ()
Created:      2004-08-20 04:34:05.881763200 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified:  2004-08-20 04:34:06.041993600 (India Standard Time)
Accessed:      2004-08-20 04:34:06.031979200 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: excel.xls
Parent MFT Entry: 346  Sequence: 348
Allocated Size: 5632  Actual Size: 5632
Created:      2004-08-20 04:34:05.881763200 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified:  2004-08-20 03:54:33.708139200 (India Standard Time)
Accessed:      2004-08-20 04:34:06.031979200 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)  Name: N/A  Resident  size: 72
Type: $FILE_NAME (48-5)  Name: N/A  Resident  size: 84
Type: $DATA (128-4)  Name: N/A  Non-Resident  size: 5632  init_size: 5632
3324115 3324116 3324117 3324118 3324119 3324120 3324121 3324122
3324123 3324124 3324125
```

Step 7: Examine the Generated File List

The generated file_list.txt file was opened to examine the files and directories identified by Sleuth Kit.



The screenshot shows a file explorer window with a dark theme. The address bar at the top displays the path: `-trriage-a | soc-trriage-a | .env | file_li |`. The main pane shows a directory tree with the following structure:

- `|-/d 3671-144-7: Documents and Settings`
 - `+ -/d 344-144-6: Mr. Evil`
 - `++ -/r 345-128-4: NTUSER.DAT`
 - `++ -/d 346-144-6: Templates`
 - `+++ r/r 9760-128-4: amipro.sam`
 - `+++ r/r 9759-128-4: excel.xls`
 - `+++ r/r 9758-128-4: excel4.xls`
 - `+++ r/r 9757-128-4: lotus.wk4`
 - `+++ r/r 9756-128-4: powerpnt.ppt`
 - `+++ r/r 9755-128-3: presenta.shw`
 - `+++ r/r 9754-128-4: quatttro.wb2`
 - `+++ r/r 9753-128-3: sndrec.wav`
 - `+++ r/r 9752-128-4: winword.doc`
 - `+++ r/r 9750-128-4: winword2.doc`
 - `+++ r/r 9751-128-3: wordpfct.wpd`
 - `+++ r/r 9749-128-3: wordpfct.wpg`
 - `++ -/d 3665-144-1: Start Menu`
 - `+++ r/r 9778-128-3: desktop.ini`
 - `+++ d/d 4156-144-5: Programs`
 - `++++ -/d 4186-144-1: Startup`
 - `+++++ r/r 9762-128-3: desktop.ini`
 - `++++ -/d 4746-144-6: Accessories`
 - `+++++ -/d 5502-144-1: Entertainment`
 - `+++++ r/r 9770-128-3: desktop.ini`
 - `+++++ -/d 5712-144-6: Accessibility`
 - `+++++ -/r 9773-128-4: Utility Manager.lnk`
 - `+++++ -/r 9774-128-4: On-Screen Keyboard.lnk`
 - `+++++ -/r 9775-128-4: Narrator.lnk`
 - `+++++ -/r 9776-128-4: Magnifier.lnk`
 - `+++++ -/r 9777-128-3: desktop.ini`
 - `+++++ -/r 9765-128-4: Windows Explorer.lnk`
 - `+++++ -/r 9766-128-4: Tour Windows XP.lnk`
 - `+++++ -/r 9767-128-4: Synchronize.lnk`
 - `+++++ -/r 9768-128-3: Program Compatibility Wizard.lnk`
 - `+++++ -/r 9769-128-4: Notepad.lnk`
 - `+++++ -/r 9771-128-3: desktop.ini`
 - `+++++ -/r 9772-128-4: Command Prompt.lnk`
 - `+++++ -/r 9845-128-4: Address Book.lnk`
 - `++++ -/r 9761-128-5: Windows Media Player.lnk`
 - `++++ -/r 9763-128-4: Remote Assistance.lnk`
 - `++++ -/r 9764-128-3: desktop.ini`

Step 8: Timeline Analysis (Optional)

The experiment manual specifies timeline analysis as an optional step. A body file was generated using fls.

The Sleuth Kit Windows package contains mactime.pl rather than mactime.exe. Perl was not available in the Windows environment, so the optional mactime timeline generation was not executed.

```
PS C:\Users\bhargav\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
Get-ChildItem -Path .. -Filter "mactime*" -Recurss
e
Directory: C:\Users\bhargav\Downloads\sleuthkit-4.14.0-win32\
sleuthkit-4.14.0-win32\bin

Mode                LastWriteTime         Length Name
----                -
-a-----         08-09-2026    09:11         27249 mactime.pl
```

Step 9: Generate and Collect the Report Data

The generated analysis files were collected

Name	Date modified
 4Dell Latitude CPi.E01	28-07-2026 10:25
 file_list.txt	08-09-2026 09:34
 filesystem_info.txt	08-09-2026 09:20
 recovered_excel.xls	08-09-2026 09:37
 body.txt	08-09-2026 09:44
 metadata_info.txt	08-09-2026 09:47
 partitions.txt	08-09-2026 09:47

Result

The forensic disk image 4Dell Latitude CPi.E01 was successfully analyzed using Sleuth Kit.

The following operations were successfully performed:

- Identified the file system as NTFS using fsstat.
- Identified the partition structure using mmls.
- Listed files and directories using fls.
- Identified the inode of excel.xls.
- Recovered excel.xls using icat.

- Analyzed file metadata using istat.
- Generated a body file for optional timeline analysis.

The recovered file was saved as: recovered_excel.xls

Conclusion

Sleuth Kit was successfully used to analyze the forensic disk image and extract digital evidence. The experiment demonstrated file-system analysis, partition identification, file listing, file recovery, and metadata analysis using command-line forensic tools.